

A soccer player tracked their goals throughout the most recent soccer season. He scored 20 goals over a total of six games. This data is outlined in the table below.

Game	Goals scored
1	6
2	4
3	3
4	1
5	2
6	4

1. Does the data show a proportional relationship between goals scored and games? Explain.

No, to be proportional it should have a $\frac{y}{x}$ constant relationship.

$$\frac{6}{1} \neq \frac{4}{2}$$

$$6 \neq 2$$

A recipe for enchiladas shows that 9 cups of pinto beans will produce 3 servings of enchiladas.

2. How many servings will 1 cup of pinto beans produce?

$$\frac{9 \text{ cups}}{3 \text{ servings}} = \frac{1 \text{ cup}}{x \text{ servings}}$$

$$9 \cdot x = 3 \cdot 1$$

$$\frac{9x}{9} = \frac{3}{9}$$

$$x = \frac{3}{9} = \frac{1}{3}$$

$$x = \frac{1}{3} \text{ of a serving}$$

3. Complete the following table.

Cups of Pinto Beans (x)	Servings (y)
1	$\frac{1}{3}$
6	$6 \cdot \frac{1}{3} = \frac{6}{3} = 2$
9	3
12	$12 \cdot \frac{1}{3} = \frac{12}{3} = 4$
15	$15 \cdot \frac{1}{3} = \frac{15}{3} = 5$

4. Write an equation that could be used to find the number of servings, y , that any number of cups of pinto beans, x , could produce.

$$y = \frac{1}{3}x$$

5. What is the constant of proportionality?

$$\frac{1}{3}$$

6. Using the same enchilada recipe, complete the following table based on servings and amount of enchiladas.

Number of enchiladas (x)	Number of servings (y)
1	$\frac{1}{2}$
4	2
16	8
24	12
30	15